Wenbo Zhi

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Education

University of Waterloo Waterloo, ON

Honours Electrical Engineering, Co-op Program *Sep 2025 – Expected 2030*

- **Relevant Courses:** ECE 106 (E&M), ECE 108 (Logic), ECE 124 (Digital Circuits), ECE 140 (Linear Circuits), ECE 150 (Programming)
- **Certifications (Coursera):** Data Structures & Algorithmic Toolbox (UCSD), Intro to Electronics (Georgia Tech)

Technical Skills

Languages: C/C++, Python, TypeScript, JavaScript, PowerShell, Bash, NSIS, VHDL
Technologies: RTOS (PREEMPT_RT), Industrial Automation, CI/CD, Digital Signal Processing, Embedded Firmware, PCB Assembly, WeChat Cloud Development (Serverless)
Developer Tools: Git/GitHub, CMake, CODESYS, LTspice, MATLAB/Simulink, Linux, LaTeX

Work Experience

Electrium Mobility Sep 2026 – Present

Web Developer (Co-op) | TypeScript, JavaScript, Python *Waterloo, ON (Remote)*

- Developing and maintaining features for the team's web platform in TypeScript and JavaScript, with Python on the backend.
- Building responsive front-end components and debugging issues across the stack for desktop and mobile.
- Collaborating remotely via Git/GitHub pull requests and code reviews, shipping updates in small increments.

Sioux Technologies Jan 2026 – Apr 2026

Control Software Intern | C++, NSIS, PowerShell, Linux, CODESYS *Suzhou, China (Hybrid)*

- Contributed to the proprietary "Smart Suite" ecosystem, directly programming and calibrating functional parameters for the "Smart Controller" to enhance hardware-software synchronization.
- Engineered an atomic rollback mechanism in NSIS and C++ to protect system integrity, driving OS-level configuration errors on production controllers down to zero.
- Refactored legacy PowerShell build scripts and implemented incremental caching strategies, cutting software build times by 75% (from 20 min to 5 min).
- Configured PREEMPT_RT Linux kernels for high-performance controllers and deployed CODESYS runtime environments to meet strict microsecond-level real-time execution constraints.

Projects

Midnight Sun Solar Race Car Jan 2026 – Present

PCB, CAN Bus, C++, Git | UWaterloo's solar vehicle racing team

- Performing extensive PCB testing, precision soldering, and complex wire harnessing to ensure robust physical integration of mission-critical vehicle electronics.
- Collaborating on a telemetry data analysis pipeline to model solar array power generation curves under dynamic conditions and inform real-time competitive racing strategy.
- Hardening Battery Management System (BMS) communication over CAN bus through interrupt-timing optimization to guarantee data integrity under extreme operating conditions.

DeployGuard CLI – Fault-Tolerant Deployment Orchestrator Present

C++17, CMake, Bash, YAML GitHub

- Built an industrial-grade C++ CLI that wraps and monitors external deployment scripts/binaries, capturing process exit codes in real time to detect failed deployments.
- Designed a YAML-driven configuration layer that decouples deployment orchestration logic from execution scripts, so recovery and rollback behavior can be customized without recompiling.
- Implemented automatic self-healing recovery in a lightweight, dependency-free C++ state machine: on a non-zero exit code or crash, the CLI halts the faulty deployment and restores the last known-good state.

Subscription Renewal Tracker Jul 2026 – Present

JavaScript, WeChat Cloud Development, Serverless Functions, NoSQL GitHub

- Independently designed and built a full-stack WeChat Mini Program that tracks subscription/membership renewals and pushes reminders before users get billed, from data modeling through backend automation.
- Architected a serverless backend on WeChat Cloud Development (cloud functions + NoSQL cloud database) with per-user data isolation and a daily scheduled function that scans upcoming renewals against configurable reminder windows.
- Designed a re-authorization UX flow around the platform's one-time subscription-message limit, and handled billing-cycle edge cases such as month-end and custom-interval rollovers.